

MPhil Thesis Circus

Maximilian Kasy

24.11.2024

Statistics, econometrics, machine learning

- Statistics as a social process:
 - Publication bias, p-hacking, pre-analysis plans, ...
 - How can we incorporate a multi-agent perspective into statistical theory?
 - Mechanism design and the foundations of statistics.
- Machine learning and econometrics:
 - Online / active learning and the design of adaptive experiments.
 - Automated decision-making and welfare economics.
 - Supervised learning, regularization, tuning, and the decision theory of shrinkage estimators.

The political economy and social impact of AI

- Who picks the objectives?
- Discrimination / impact on inequality.
- Value alignment.
- Explainability of automated decisions.
- Privacy and data governance.

Forthcoming book:

The means of prediction: How AI works, and for whom

Field experiments and policy pilots

- Basic income.
- Job guarantee programs.
- Refugee relocation.
- Environmental externalities and taxation.
- Taking adaptive methods to the field.

Big field experiment that hopefully starts next year:

A democratic basic income